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LOW TEMPERATURE ATMOSPHERIC EPITAXIAL PROCESS

Abstract

A method of epitaxial deposition is disclosed. The method includes plasma pre-cleaning a substrate having openings with an aspect ratio of greater than 5:1. The plasma pre-cleaning is performed at pre-clean pressure of less than 5 Torr. The method also includes, after the plasma pre-cleaning, depositing a material in the openings using epitaxial deposition at a deposition pressure of about 700 Torr to about 800 Torr. In another embodiment, a method of gap filling using epitaxial deposition includes patterning a substrate with openings having an aspect ratio of 5:1. The method also includes plasma pre-cleaning the substrate, the plasma pre-cleaning being performed at a pre-clean pressure of about 1 Torr to about 5 Torr. The method also includes, after the plasma pre-cleaning, depositing a gap fill material in the openings using epitaxial deposition at a deposition pressure of about 700 Torr to about 800 Torr.

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Background/Summary

BACKGROUND

Field

[0001] Embodiments described herein generally relate to semiconductor device fabrication. More specifically, embodiments of the present disclosure relate to methods for epitaxial deposition.

Description of the Related Art

[0002] In fabrication of an integrated circuit, prior to the epitaxial deposition process, a high temperature bake at a temperature of greater than 1100° C. is usually utilized to remove a native oxide. However, the high temperature bake may cause issues such as dopant diffusion and silicon migration. Further, the high temperature bake may damage the hard mask of a patterned substrate, which may prevent realization of selective atmospheric epitaxial processes.

[0003] In order to overcome these drawbacks, low pressure epitaxial deposition has been implemented. However, low pressure epitaxial deposition may have low throughput in a manufacturing environment due to the low growth rate of the deposited film.

[0004] Therefore, there is a need for more efficient methods of epitaxial deposition with dopant diffusion and silicon migration control.

SUMMARY

[0005] Embodiments of the present disclosure provide a method of epitaxial deposition. The method includes plasma pre-cleaning a substrate having openings with an aspect ratio of greater than 5:1. The plasma pre-cleaning is performed at pre-clean pressure of less than 5 Torr. The method also includes, after the plasma pre-cleaning, depositing a material in the openings using epitaxial deposition at a deposition pressure of about 700 Torr to about 800 Torr.

[0006] Embodiments of the present disclosure provide a method of gap filling using epitaxial deposition. The method includes using epitaxial deposition includes patterning a substrate with openings having an aspect ratio of 5:1. The method also includes plasma pre-cleaning the substrate, the plasma pre-cleaning being performed at a pre-clean pressure of about 1 Torr to about 5 Torr. The method also includes, after the plasma pre-cleaning, depositing a gap fill material in the openings using epitaxial deposition at a deposition pressure of about 700 Torr to about 800 Torr.

[0007] Embodiments of the present disclosure provide a processing system. The processing system includes one or more processing chambers and a system controller configured to cause the processing system to perform, in the one or more processing chambers, a method. The method includes plasma pre-cleaning a substrate having openings with an aspect ratio of greater than 5:1. The plasma pre-cleaning is performed at a pre-clean pressure of about 1 Torr to about 5 Torr. The method also includes, after the plasma pre-cleaning, depositing a material in between the structures using epitaxial deposition at a deposition pressure of about 700 Torr to about 800 Torr.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] So that the manner in which the above recited features of embodiments of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical

embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

[0009] FIG. 1 illustrates a schematic top view of a multi-chamber processing system, according to embodiments of the present disclosure.

[0010] FIG. 2 is a cross-sectional view of the pre-clean system from the multi-chamber processing system of FIG. 1, according to one embodiment.

[0011] FIG. 3 illustrates a method for low temperature epitaxial deposition, according to embodiments of the present disclosure.

[0012] FIG. 4 illustrates a method for low temperature gap filling using epitaxial deposition, according to embodiments of the present disclosure.

[0013] FIG. 5 is a cross-sectional view of the substrate having structures 507 disposed thereon, according to embodiments.

[0014] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation. It is contemplated that elements disclosed in some embodiments may be beneficially utilized on other implementations without specific recitation.

DETAILED DESCRIPTION

[0015] Embodiments described herein generally relate to semiconductor device fabrication. More specifically, embodiments of the present disclosure relate to methods for epitaxial deposition.

[0016] FIG. 1 is a schematic top view of a multi-chamber processing system 100, according to one or more embodiments of the present disclosure. The multi-chamber processing system 100 generally includes a factory interface 102, load lock chambers 104, 106, transfer chambers 108, 110 with respective transfer robots 112, 114, holding chambers 116, 118, and processing chambers 120, 122, 124, 126, 128, 130. As detailed herein, substrates in the multi-chamber processing system 100 can be processed in and transferred between the various chambers without exposing the substrates to an ambient environment exterior to the processing system 100. For example, the substrates can be processed in and transferred between the various chambers maintained at a low pressure (e.g., less than or equal to about 300 Torr) or vacuum environment without breaking the low pressure or vacuum environment among various processes performed on the substrates in the processing system 100. Accordingly, the multi-chamber processing system 100 may provide for an integrated solution for some processing of substrates.

[0017] In the illustrated example of FIG. 1, the factory interface 102 includes a docking station 132 and factory interface robots 134 to facilitate transfer of substrates. The docking station 132 is adapted to accept one or more front opening unified pods (FOUPs) 136. In some examples, each factory interface robot 134 generally includes a blade 138 disposed on one end of the respective factory interface robot 134 adapted to transfer the substrates from the factory interface 102 to the load lock chambers 104, 106.

[0018] The load lock chambers 104, 106 have respective ports 140, 142 coupled to the factory interface 102 and respective ports 144, 146 coupled to the transfer chamber 108. The transfer chamber 108 further has respective ports 148, 150 coupled to the holding chambers 116, 118 and respective ports 152, 154 coupled to processing chambers 120, 122. Similarly, the transfer chamber 110 has respective ports 156, 158 coupled to the holding chambers 116, 118 and respective ports 160, 162, 164, 166 coupled to processing chambers 124, 126, 128, 130. The ports 144, 146, 148, 150, 152, 154, 156, 158, 160, 162, 164, 166 can be, for example, slit valve openings with slit valves for passing substrates therethrough by the transfer robots 112, 114 and for providing a seal between respective chambers to prevent a gas from passing between the respective chambers. Generally, any port is open for transferring a substrate therethrough. Otherwise, the port is closed.

[0019] The load lock chambers 104, 106, transfer chambers 108, 110, holding chambers 116, 118, and processing chambers 120, 122, 124, 126, 128, 130 may be fluidly coupled to a gas and pressure

control system. The gas and pressure control system can include one or more gas pumps (e.g., turbo pumps, cryo-pumps, roughing pumps), gas sources, various valves, and conduits fluidly coupled to the various chambers. In operation, a factory interface robot **134** transfers a substrate from a FOUP **136** through a port **140** or **142** to a load lock chamber **104** or **106**. The gas and pressure control system then pumps down the load lock chamber **104** or **106**. The gas and pressure control system further maintains the transfer chambers **108**, **110** and holding chambers **116**, **118** with an interior low pressure or vacuum environment (which may include an inert gas). Hence, the pumping down of the load lock chamber **104** or **106** facilitates passing the substrate between, for example, the atmospheric environment of the factory interface **102** and the low pressure or vacuum environment of the transfer chamber **108**.

[0020] With the substrate in the load lock chamber **104** or **106** that has been pumped down, the transfer robot **112** transfers the substrate from the load lock chamber **104** or **106** into the transfer chamber **108** through the port **144** or **146**. The transfer robot **112** is then capable of transferring the substrate to and/or between any of the processing chambers **120**, **122** through the respective ports **152**, **154** for processing and the holding chambers **116**, **118** through the respective ports **148**, **150** for holding to await further transfer. Similarly, the transfer robot **114** is capable of accessing the substrate in the holding chamber **116** or **118** through the port **156** or **158** and is capable of transferring the substrate to and/or between any of the processing chambers **124**, **126**, **128**, **130** through the respective ports **160**, **162**, **164**, **166** for processing and the holding chambers **116**, **118** through the respective ports **156**, **158** for holding to await further transfer. The transfer and holding of the substrate within and among the various chambers can be in the low pressure or vacuum environment provided by the gas and pressure control system.

[0021] The processing chambers **120**, **122**, **124**, **126**, **128**, **130** can be any appropriate chamber for processing a substrate. In some examples, the processing chamber **120** can be capable of performing an etch process, the processing chamber **122** can be capable of performing a cleaning process, and the processing chambers **124**, **126**, **128**, **130** can be capable of performing respective deposition processes.

[0022] A system controller **168** is coupled to the multi-chamber processing system **100** for controlling the multi-chamber processing system **100** or components thereof. For example, the system controller **168** may control the operation of the multi-chamber processing system **100** using a direct control of the chambers **104**, **106**, **108**, **110**, **116**, **118**, **120**, **122**, **124**, **126**, **128**, **130** of the processing system multi-chamber **100** or by controlling controllers associated with the chambers **104**, **106**, **108**, **110**, **116**, **118**, **120**, **122**, **124**, **126**, **128**, **130**. In operation, the system controller **168** enables data collection and feedback from the respective chambers to coordinate performance of the processing system **100**.

[0023] The system controller **168** generally includes a central processing unit (CPU) **170**, memory **172**, and support circuits **174**. The CPU **170** may be one of any form of a general-purpose processor that can be used in an industrial setting. The memory **172**, or non-transitory computer-readable medium, is accessible by the CPU **170** and may be one or more of memory such as random access memory (RAM), read only memory (ROM), floppy disk, hard disk, or any other form of digital storage, local or remote. The support circuits **174** are coupled to the CPU **170** and may comprise cache, clock circuits, input/output subsystems, power supplies, and the like. The various methods disclosed herein may generally be implemented under the control of the CPU **170** by the CPU **170** executing computer instruction code stored in the memory **172** (or in memory of a particular processing chamber) as, for example, a software routine. When the computer instruction code is executed by the CPU **170**, the CPU **170** controls the chambers to perform processes in accordance with the various methods.

[0024] Other processing systems can be in other configurations. For example, more or fewer processing chambers may be coupled to a transfer apparatus. In the illustrated example, the transfer apparatus includes the transfer chambers **108**, **110** and the holding chambers **116**, **118**. In other

examples, more or fewer transfer chambers (e.g., one transfer chamber) and/or more or fewer holding chambers (e.g., no holding chambers) may be implemented as a transfer apparatus in a processing system.

[0025] FIG. 2 is a cross-sectional view of the pre-clean system **200**. The pre-clean system **200** may be one or more of the processing chambers **120**, **122**, **124**, **126**, **128**, **130**. The pre-clean system **200** includes a pre-clean chamber **201** (also referred to as a process chamber). The pre-clean chamber **201** includes a chamber body **210**. The chamber body **210** includes a bottom **211**, a lid assembly **214**, and one or more chamber walls **212** connecting the bottom **211** with the lid assembly **214**. The chamber body **210** can enclose an interior volume **205** of the pre-clean chamber **201**.

[0026] The pre-clean chamber **201** further includes a substrate support assembly **216**. The substrate support assembly **216** can include a substrate support **232**, an actuator **234**, and a shaft **236** connecting the actuator **234** with the substrate support **232**. The substrate support **232** can be located in the interior volume **205** to support a substrate **50** during processing.

[0027] The chamber body **210** can further include a slit valve **215** to allow insertion and removal of a substrate **50** into and from the interior volume **205** of the pre-clean chamber **201**. The pre-clean system **200** and multi-chamber processing system **100** can be configured to have a pressure in the interior volume **205** remain below a pressure in the transfer chamber **108** when the slit valve **215** is opened to prevent flow of gas and/or particles from the pre-clean chamber **201** to the transfer chamber **108** as described in further detail below.

[0028] The lid assembly **214** is disposed at an upper end of the chamber body **210**. The lid assembly **214** can include a remote plasma source **220** for generating a plasma from cleaning gases provided to the remote plasma source **220**. The cleaning gases can be provided from a cleaning gas source **227** through a gas inlet **226** of the pre-clean chamber **201**. The cleaning gas source **227** can include a separate tank for each cleaning gas. In one embodiment, the cleaning gases from the cleaning gas source **227** can include one or more of hydrogen (H_{2}), nitrogen trifluoride (NF_{3}), and ammonia (NH_{3}). The remote plasma source **220** can include a first electrode **221** and a second electrode **222**. The first electrode **221** can be spaced apart from the second electrode **222**. The remote plasma source **220** can include a plasma-generating volume **229** positioned between the first electrode **221** and the second electrode **222**.

[0029] The pre-clean system **200** can include a radio frequency (RF) power source **224**. The RF power source **224** can be connected to the first electrode **221**. The second electrode **222** can be connected to electrical ground to serve as a return path for the RF power when the plasma is generated in the volume **229**. The RF power source **224** can be used to generate a plasma of the cleaning gases inside plasma-generating volume **229** when the cleaning gases are provided to the remote plasma source **220**.

[0030] The lid assembly **214** can further include a blocker plate **228** and a showerhead **230** for distributing gas and/or plasma to the interior volume **205** of the pre-clean chamber **201**. The blocker plate **228** can be positioned between the remote plasma source **220** and the showerhead **230**. The blocker plate **228** can receive plasma and/or gas discharged from the remote plasma source **220**. In some embodiments, one or more gases may be provided directly to the blocker plate **228** or showerhead **230** allowing the remote plasma source **220** to be bypassed.

[0031] The pre-clean system **200** can further include an inert gas source **240** connected to the pre-clean chamber **201**. In one embodiment, the inert gas source **240** includes nitrogen, but in other inert gases (e.g., argon) may also be used. The inert gas can be used to pressurize the interior volume **205** of the pre-clean chamber **201** after a pre-clean process is performed on the substrate **50** and/or before a new substrate **50** is transferred into the pre-clean chamber **201**. The pre-clean system **200** can include a pressure sensor **260** configured to measure a pressure of the interior volume **205** of the pre-clean chamber **201**.

[0032] The inert gas source **240** can be connected to the gas inlet **226** of the process chamber through a first supply line **245** or a second supply line **246** of the pre-clean system **200**. The first

supply line **245** and the second supply line **246** can be connected to the gas inlet **226** through a common supply line **247**. The first supply line **245** and the second supply line **246** can be arranged to form parallel (i.e., alternative) paths relative to each other, so that gas can be supplied to the pre-clean chamber **201** through one of the supply lines without going through the other supply line. [0033] The first supply line **245** can include a first supply valve **241** that can be opened to connect the first supply line **245** with the common supply line **247**. The second supply line **246** can include a second supply valve **242** that can be opened to connect the second supply line **246** with the common supply line **247**.

[0034] The first supply line **245** can have a smaller internal diameter relative to the internal diameter of the second supply line **246**. In some embodiments, the internal diameter of the first supply line **245** can be from about 5% to about 90%, such as from about 10% to about 50% of the internal diameter of the second supply line **246**. The smaller diameter of the first supply line **245** can be used to slowly raise the pressure in the interior volume **205** from the vacuum pressures (e.g., 2-20 Torr, such as between about 3-5 Torr) used for the pre-clean process after a pre-clean process is performed on the substrate **50**. On the other hand, the second supply line **246** can be used to quickly raise the pressure in the interior volume **205** back to atmospheric pressure or a pressure near atmospheric pressure after the pressure reaches a higher pressure (e.g., 300 Torr) from the gas provided from the smaller first supply line **245**. Slowly raising the pressure after the pre-clean process can prevent the likelihood of damaging the substrate **50** from an abrupt pressure change, such as mechanical damage caused by a wobbling or otherwise unintentionally moving the substrate **50**.

[0035] Using different supply lines with different internal diameters is one method of varying the rate at which gas is provided to the interior volume **205**. In other embodiments, the slower pressure changes can be achieved, for example, with an analog control valve on a single supply line. In some of these other embodiments, a sensor, such as a flowmeter or pressure sensor can be used to control the analog control valve or other actuator (e.g., a variable-speed pump) in order to control the rate at which the pressure in the interior volume **205** increases when the inert gas is supplied to the interior volume **205**, so that slower pressure changes in the interior volume **205** can be achieved.

[0036] The pre-clean system **200** can further include a vacuum pump **218** configured to exhaust gas from the pre-clean chamber **201** through an exhaust port **223** of the pre-clean chamber **201**. The vacuum pump **218** can be connected to the exhaust port **223** through a first exhaust line **261** or a second exhaust line **262** of the pre-clean system **200**. The first exhaust line **261** and the second exhaust line **262** can be arranged to form parallel (i.e., alternative) paths relative to each other, so that gas can be exhausted from the pre-clean chamber **201** through one of the exhaust lines without going through the other exhaust line. The first exhaust line **261** and the second exhaust line **262** can be connected to the exhaust port **223** through a common exhaust line **263**. The first exhaust line **261** can include a first exhaust valve **219** that can be opened to fluidly couple the first exhaust line **261** with the common exhaust line **263**. The second exhaust line **262** can include a second exhaust valve **239** that can be opened to fluidly couple the second exhaust line **262** with the common exhaust line **263**.

[0037] The first exhaust line **261** can have a smaller internal diameter relative to the internal diameter of the second exhaust line **262**. All references provided in this disclosure to internal diameters also apply to internal cross-sectional areas, for example if the component (e.g., a fluid conduit) has a non-circular cross-section. In some embodiments, the internal diameter of the first exhaust line **261** can be from about 5% to about 75%, such as from about 10% to about 50% of the internal diameter of the second exhaust line **262**. The smaller diameter of the first exhaust line **261** can be used to smoothly and slowly lower the pressure in the interior volume **205** from atmospheric pressure or a pressure near atmospheric pressure (e.g., 700-800 Torr), to a lower pressure, such as from about 400-650 Torr, such as about 600 Torr. The pressure reduction can be performed, for

example, after a substrate **50** is transferred into the pre-clean chamber **201** from the transfer chamber **108**, which is maintained at atmospheric pressure or a pressure near atmospheric pressure. On the other hand, the second exhaust line **262** can be used to quickly lower the pressure in the interior volume **205** down to a pressure near the pressure used for the pre-clean plasma process, such as a pressure less than 50 Torr, such as about 100 mTorr to about 20 Torr, such as a pressure between about 300 mTorr and about 5 Torr). Slowly lowering the pressure after a substrate **50** is transferred into the pre-clean chamber **201** can prevent the likelihood of damaging the substrate **50** from an abrupt pressure change, such as mechanical damage caused by wobbling or otherwise unintentionally moving the substrate **50**.

[0038] Using different exhaust lines with different internal diameters is one method of varying the rate at which gas and/or plasma is exhausted from the interior volume **205**, so that slower pressure changes in the interior volume **205** can be achieved. In other embodiments, the slower pressure changes can be achieved, for example, with an analog control valve on a single exhaust line. In some of these other embodiments, a sensor, such as a flowmeter or pressure sensor can be used to control the analog control valve or other actuator (e.g., a variable-speed vacuum pump) in order to control the rate at which the pressure in the interior volume **205** decreases when the interior volume **205** is brought down to a vacuum pressure for performing the plasma pre-clean process.

[0039] As introduced above, the substrate support assembly **216** includes the substrate support **232**, the actuator **234**, and the shaft **236** connecting the actuator **234** with the substrate support **232**. The shaft **236** can extend through a centrally-located opening formed in the bottom **211** of the chamber body **210**. The actuator **234** may be flexibly sealed to the bottom **211** of the chamber body **210** by bellows (not shown) that prevent vacuum leakage from around the shaft **236**. The actuator **234** allows the substrate support **232** to be moved vertically within the chamber body **210** between a process position and a lower transfer position. The transfer position can be slightly below the opening of the slit valve **215** formed through one of the one or more walls **212** of the chamber body **210**.

[0040] Although not shown, in some embodiments, an RF and/or DC bias can be coupled to the substrate support **232** to assist with directing the cleaning plasma toward the substrate **50**.

[0041] The pre-clean system **200** can further include an auxiliary exhaust assembly **270**. The auxiliary exhaust assembly **270** can include a first auxiliary exhaust line **275**, a second auxiliary exhaust line **276**, and a common auxiliary exhaust line **278**. The auxiliary exhaust assembly **270** can further include a vacuum pump or other device for creating a negative pressure in the auxiliary exhaust assembly **270** lines relative to the interior volume **205** of pre-clean chamber **201**, so that gas is exhausted from the interior volume **205** through the auxiliary exhaust assembly **270** when the valves of the auxiliary exhaust assembly **270** are opened.

[0042] The common auxiliary exhaust line **278** can be connected to the interior volume **205** of the pre-clean chamber **201**. The first auxiliary exhaust line **275** and the second auxiliary exhaust line **276** can be connected to the interior volume **205** of the pre-clean chamber **201** through the common auxiliary exhaust line **278**. The first auxiliary exhaust line **275** can include a first auxiliary exhaust valve **272** that can be opened to connect the first auxiliary exhaust line **275** with the common auxiliary exhaust line **278**. The second auxiliary exhaust line **276** can include a second auxiliary exhaust valve **274** that can be opened to connect the second auxiliary exhaust line **276** with the common auxiliary exhaust line **278**.

[0043] The first auxiliary exhaust valve **272** can be opened when a high pressure condition occurs. The first auxiliary exhaust line **275** can include a pressure sensor **271** to measure a pressure inside the first auxiliary exhaust line **275**. Upon measuring a pressure above a given threshold (e.g., 800 Torr), the first auxiliary exhaust valve **272** can be opened to relieve pressure inside the interior volume **205**. Because the pre-clean chamber **201** is operated at a higher pressure than other pre-clean chambers that typically operate at vacuum pressures (e.g., less than 100 Torr) for the pre-clean process and substrate transfer, more components of the pre-clean chamber are fastened or

otherwise secured to each other. For example, the components of the lid assembly **214** can be secured to other components in the lid assembly **214** and/or to the chamber walls **212**. Some of these components in the lid assembly are generally unfastened for pre-clean chambers that operate at vacuum pressures for the pre-clean process and substrate transfer to and from the pre-clean chamber. The additional fastening of components in the pre-clean chamber **201** can help prevent movement of any of the components during the pressure changes that occur for each substrate pre-clean and transfer as described in further detail below. Securing these components though can create a safety issue as the previously unsecured components could move to relieve a high pressure situation. In the pre-clean chamber **201**, the first auxiliary exhaust valve **272** can open to relieve a high pressure condition when measured by the pressure sensor **271** and prevent an unsafe high-pressure condition from occurring.

[0044] The second auxiliary exhaust valve **274** can be opened when the slit valve **215** is opened, which allows gas to flow from the interior volume **205** and out the auxiliary exhaust assembly **270**. The interior volume **205** of the pre-clean chamber **201** is generally considered to be less clean than the interior volume of the transfer chamber **108**. Thus, gas should not flow from the interior volume **205** of the pre-clean chamber **201** to the interior volume of the transfer chamber **108**. Opening the second auxiliary exhaust valve **274** when the slit valve **215** opens reduces the pressure in the interior volume **205** relative to the pressure in the interior volume of the transfer chamber **108** and gas flows from the interior volume of the transfer chamber **108** through the interior volume **205** of the pre-clean chamber **201** and out through the auxiliary exhaust assembly **270**.

[0045] The pre-clean system **200** can also include a controller **290** for controlling processes within the pre-clean system **200** (FIG. 2) and other portions of the processing system **100** (FIG. 1). The controller **290** can be any type of controller used in an industrial setting, such as a programmable logic controller (PLC). The controller **290** includes a processor **292**, a memory **294**, and input/output (I/O) circuits **296**. The controller **290** can further include one or more of the following components (not shown), such as one or more power supplies, clocks, communication components (e.g., network interface card), and user interfaces typically found in controllers for semiconductor equipment.

[0046] The memory **294** can include non-transitory memory. The non-transitory memory can be used to store the programs and settings described below. The memory **294** can include one or more readily available types of memory, such as read only memory (ROM) (e.g., electrically erasable programmable read-only memory (EEPROM), flash memory, floppy disk, hard disk, or random access memory (RAM) (e.g., non-volatile random access memory (NVRAM).

[0047] The processor **292** is configured to execute various programs stored in the memory **294**, such as a program configured to execute the method **1000** described below in reference to FIG. 3. During execution of these programs, the controller **290** can communicate to I/O devices (e.g., sensors and actuators) through the I/O circuits **296**. For example, during execution of these programs and communication through the I/O circuits, the controller **290** can control outputs (e.g., open and close valves) and receive information from feedback devices (e.g., feedback on the open/close state of valves), sensors, and other instrumentation in the pre-clean system **200** and other portions of the multi-chamber processing system **100**.

[0048] The memory **294** can further include various operational settings used to control the pre-clean system **200** and other portions of the multi-chamber processing system **100**. For example, the settings can include pressure settings for when a transition between slowly changing and more quickly changing the pressure in the interior volume **205** is made in the method **1000** as described below in reference to FIG. 3 among various other settings.

[0049] FIG. 3 is a method **300** for low temperature epitaxial deposition. At operation **301**, a substrate is disposed in the factory interface **102**. The factory interface **102** includes a docking station **132** and factory interface robots **134** to facilitate transfer of substrates. The docking station **132** is adapted to accept one or more front opening unified pods (FOUPs) **136**.

[0050] At operations **302**, the substrates are transferred from the factory interface **102** to the load lock chambers **104, 106**. In some examples, each factory interface robot **134** generally includes a blade **138** disposed on one end of a respective factory interface robot **134** adapted to transfer the substrates from the factory interface **102** to the load lock chambers **104, 106**. The load lock chambers **104, 106** have respective ports **140, 142** coupled to the factory interface **102** and respective ports **144, 146** coupled to a transfer chamber **108**.

[0051] At operation **303**, the substrate is transferred from the load lock chambers **104, 106** to the transfer chamber **108**. The transfer chamber **108** has respective ports **148, 150** coupled to the holding chambers **116, 118** and respective ports **152, 154** coupled to processing chambers **120, 122**.

[0052] At operation **304**, the substrate is transferred from the transfer chamber **108** to the processing chambers **120, 122**. The transfer chamber **108** has a pressure of about 700 Torr to about 800 Torr, such as about 760 Torr.

[0053] At operation **305**, the substrate is plasma pre-cleaned in the processing chambers **120, 122**. The pre-clean process includes removing a native-oxide from the substrate. The plasma pre-clean is conducted at a temperature of about 100° C. to about 300° C., such as about 150° C. to about 250° C., such as about 200° C. The plasma pre-clean is conducted at a pressure of about 1 Torr to about 5 Torr, such as about 3 Torr. The plasma pre-clean is conducted for a time of about 15 second to about 45 second, such as about 30 seconds.

[0054] At operation **306**, the substrate is transferred from the processing chambers **120, 122** to the transfer chamber **110**. The substrate is transferred to the transfer chamber **110** via the holding chambers **116, 118**. The transfer chamber **110** has respective ports **156, 158** coupled to the holding chambers **116, 118** and respective ports **160, 162, 164, 166** coupled to processing chambers **124, 126, 128, 130**.

[0055] At operation **307**, the substrate is transferred from the transfer chamber to the processing chambers **124, 126, 128, 130**. The transfer chamber **108** has a pressure of about 700 Torr to about 800 Torr, such as about 760 Torr.

[0056] At operation **308**, a film is deposited on the substrate through epitaxial deposition. In some embodiments, the epitaxial deposition is a gap fill deposition, wherein the film (e.g., the deposition material) is deposited in between a plurality of structures disposed over a surface of the substrate. The structures have a high aspect ratio greater than 5:1, such as greater than 10:1. The epitaxial deposition includes deposition of the film at a temperature of about 400° C. to about 1200° C., such as about 800° C. to about 1000° C., such as about 950° C. The film is deposited using a precursor gas. The precursor gas may include a silicon precursor, such as trichlorosilane (SiHCl₃) and dichlorosilane (DCS). The epitaxial deposition is conducted at a pressure of about 700 Torr to about 800 Torr, such as about 760 Torr. The epitaxial deposition is conducted for a time of about 15 second to about 45 second, such as about 30 seconds, or about 100 seconds to about 5000 seconds, such as about 500 seconds to about 1000 seconds, such as about 700 seconds.

[0057] The atmospheric pressure (ATM) low temperature epitaxial deposition increases the dopant diffusion control, e.g., reduces the dopant diffusion into the deposited film. Pre-cleaning the substrate prior to the low temperature epitaxial deposition removes the native oxide layer from the substrate. Removal of the native oxide layer, thus, enables the low temperature epitaxial deposition. In addition, the low temperature atmospheric epitaxial deposition increases throughput in a manufacturing environment compared to low pressure epitaxial deposition. The throughput is increased via an increase in epitaxial growth during the epitaxial deposition. Further, the low temperature epitaxial deposition enables selective epitaxial deposition.

[0058] FIG. **4** is a method **400** for low temperature gap filling using epitaxial deposition. At operation **301**, a substrate is disposed in the factory interface **102**. The factory interface **102** includes a docking station **132** and factory interface robots **134** to facilitate transfer of substrates. The docking station **132** is adapted to accept one or more front opening unified pods (FOUPs) **136**.

[0059] At operations **402**, the substrates are transferred from the factory interface **102** to the load

lock chambers **104, 106**. In some examples, each factory interface robot **134** generally includes a blade **138** disposed on one end of a respective factory interface robot **134** adapted to transfer the substrates from the factory interface **102** to the load lock chambers **104, 106**. The load lock chambers **104, 106** have respective ports **140, 142** coupled to the factory interface **102** and respective ports **144, 146** coupled to a transfer chamber **108**.

[0060] At operation **403**, the substrate is transferred from the load lock chambers **104, 106** to the transfer chamber **108**. The transfer chamber **108** has respective ports **148, 150** coupled to the holding chambers **116, 118** and respective ports **152, 154** coupled to processing chambers **120, 122**.

[0061] At operation **404**, the substrate is transferred from the transfer chamber **108** to the processing chambers **120, 122**. The transfer chamber **108** has a pressure of about 700 Torr to about 800 Torr, such as about 760 Torr.

[0062] At operation **405**, a substrate is patterned in the processing chamber **120, 122**. The substrate is patterned to form high aspect ratio structures, such as structures with an aspect ratio greater than about 5:1, such as greater than about 10:1. In some embodiments, the substrate may be patterned using a lithography process to form a plurality of structures. In other embodiments, the substrate may be patterned using a deposition process to form a plurality of structures. The deposition process may include atomic layer deposition (ALD), chemical vapor deposition (CVD), or physical vapor deposition (PVD).

[0063] At operation **406**, the substrate is plasma pre-cleaned in the processing chambers **120, 122**. In some embodiments, the substrate is transferred to the other of the processing chamber **120, 122**. In other embodiments, the substrate is plasma pre-cleaned in the same processing chamber **120, 122** as the patterning. The pre-clean process includes removing a native-oxide from the substrate. The plasma pre-clean is conducted at a temperature of about 100° C. to about 300° C., such as about 150° C. to about 250° C., such as about 200° C. The plasma pre-clean is conducted at a pressure of about 1 Torr to about 5 Torr, such as about 3 Torr. The plasma pre-clean is conducted for a time of about 15 second to about 45 second, such as about 30 seconds.

[0064] At operation **407**, the substrate is transferred from the processing chambers **120, 122** to the transfer chamber **110**. The substrate is transferred to the transfer chamber **110** via the holding chambers **116, 118**. The transfer chamber **110** has respective ports **156, 158** coupled to the holding chambers **116, 118** and respective ports **160, 162, 164, 166** coupled to processing chambers **124, 126, 128, 130**.

[0065] At operation **408**, the substrate is transferred from the transfer chamber to the processing chambers **124, 126, 128, 130**. The transfer chamber **108** has a pressure of about 700 Torr to about 800 Torr, such as about 760 Torr.

[0066] At operation **409**, a gap fill material is deposited on the substrate through epitaxial deposition. The gap fill material is deposited in between the structures formed on the substrate. The epitaxial deposition includes deposition of the gap fill material at a temperature of about 400° C. to about 1200° C., such as about 800° C. to about 1000° C., such as about 950° C. The gap fill is deposited using a precursor gas. The precursor gas includes a silicon precursor, such as SiHCl.sub.3 and DCS. The flow rate of the precursor gas may depend on the precursor gas being used. For example, a flow rate for SiHCl.sub.3 a may be about 10 standard liter per minute (slm) to about 20 slm, such as about 15 slm. In another example, a flow rate for DCS may be about 400 standard cubic centimeters per minute (sccm) to about 1000 sccm. The epitaxial deposition is conducted at a pressure of about 700 Torr to about 800 Torr, such as about 760 Torr. The epitaxial deposition is conducted for a time of about 100 seconds to about 5000 seconds, such as about 500 seconds to about 1000 seconds, such as about 700 seconds.

[0067] The atmospheric pressure (ATM) low temperature epitaxial deposition increases throughput in a manufacturing environment compared to low pressure epitaxial deposition. The throughput is increased via an increase in epitaxial growth during the epitaxial deposition. Further, the low temperature epitaxial deposition enables selective epitaxial deposition, e.g., the orientation of

epitaxial deposition of the gap fill material can be selected to form a void free substrate. The low temperature epitaxial deposition enables the selective epitaxial deposition along the gaps formed in between the structures while reducing deposition across the structures. The reduction in the deposition across the structures reduces the likelihood that the gap fill material prematurely closes the gap in between the structures, enabling more gap fill material to be deposited in the gaps between the structures.

[0068] FIG. 5 is a cross-sectional view of the substrate 50 having structures 507 disposed thereon. The structures 507 define openings 508. The openings 508 are about 1.0 μm and about 1.3 μm wide, have a high aspect ratio (e.g., greater than about 5:1, such as greater than about 10:1), and a pitch between adjacent openings 508 of between about 50 nm and about 180 nm. During methods 300 and 400, the material may be deposited over the structures 507 and in the openings 508 to fill the area in between the structures 507. The structures 507 may include refractory metals, such as tungsten (W), molybdenum (Mo), tantalum (Ta), titanium (Ti), hafnium (Hf), vanadium (V), chromium (Cr), manganese (Mn), ruthenium (Ru), alloys thereof, silicide compounds thereof, nitride compounds thereof, or combinations thereof. In other examples, the structure 507 may be other metals, such as copper (Cu), nickel (Ni), cobalt (Co), iron (Fe), aluminum (Al), palladium (Pd), gold (Au), silver (Ag), platinum (Pt), alloys thereof, nitride compounds thereof, or combinations thereof. In other embodiments, the structures may be silicon or silicon germanium (SiGe).

[0069] In summary, the methods described above include low temperature epitaxial deposition processes. The low temperature atmospheric epitaxial deposition includes a deposition temperature of about 400° C. to about 1200° C. and a deposition time of about 15 seconds to about 45 seconds. The low temperature atmospheric epitaxial deposition overcomes the drawbacks of low pressure deposition by increasing throughput in a manufacturing environment. Further, the low temperature epitaxial deposition enables more efficient gap filling, selective epitaxial deposition, and increased dopant diffusion control.

[0070] While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A method of epitaxial deposition, comprising: plasma pre-cleaning a substrate having openings with an aspect ratio of greater than 5:1, wherein the plasma pre-cleaning is performed at pre-clean pressure of less than 5 Torr; and after the plasma pre-cleaning, depositing a material in the openings using epitaxial deposition at a deposition pressure of about 700 Torr to about 800 Torr.
2. The method of claim 1, further comprising: disposing the substrate in a factory interface; and transferring the substrate from the factory interface to the processing chamber.
3. The method of claim 1, wherein: the plasma pre-cleaning occurs in a first processing chamber of a substrate processing system; and the epitaxial deposition occurs in a second processing chamber of the substrate processing system.
4. The method of claim 3, further comprising: transferring the substrate from the first processing chamber to the second processing chamber.
5. The method of claim 4, wherein a transfer chamber between the first processing chamber and the second processing chamber has a pressure of about 700 Torr to about 800 Torr.
6. The method of claim 1, wherein: the plasma pre-cleaning comprises a pre-clean temperature of about 100° C. to about 300° C.; the deposition of the material comprises a deposition temperature of about 400° C. to about 1200° C.; and the deposition of the material comprises a deposition time of about 100 seconds to about 5000 seconds.
7. A method of gap filling using epitaxial deposition, comprising: patterning a substrate with

openings having an aspect ratio of 5:1; plasma pre-cleaning the substrate, wherein the plasma pre-cleaning is performed at a pre-clean pressure of about 1 Torr to about 5 Torr; and after the plasma pre-cleaning, depositing a gap fill material in the openings using epitaxial deposition at a deposition pressure of about 700 Torr to about 800 Torr.

8. The method of claim 7, further comprising: disposing the substrate in a factory interface; and transferring the substrate from the factory interface to the processing chamber.

9. The method of claim 7, wherein: the patterning occurs in a first processing chamber of a substrate processing system; the plasma pre-cleaning occurs in a second processing chamber of the substrate processing system; and the epitaxial deposition occurs in a third processing chamber of the substrate processing system.

10. The method of claim 9, further comprising: transferring the substrate from the first processing chamber to the second processing chamber; and transferring the substrate from the second processing chamber to the third processing chamber.

11. The method of claim 10, wherein a transfer chamber between the first processing chamber, the second processing chamber, and the third processing chamber has a pressure of about 700 Torr to about 800 Torr.

12. The method of claim 7, wherein: the plasma pre-clean comprises a pre-cleaning temperature of about 100° C. to about 300° C.; the deposition of the material comprises a deposition temperature of about 400° C. to about 1200° C.; and the deposition of the material comprises a deposition time of about 100 seconds to about 5000 seconds.

13. A processing system, comprising: one or more processing chambers; and a system controller configured to cause the processing system to perform, in the one or more processing chambers: plasma pre-cleaning a substrate having openings with an aspect ratio of greater than 5:1, wherein the plasma pre-cleaning is performed at a pre-clean pressure of about 1 Torr to about 5 Torr; and after the plasma pre-cleaning, depositing a material in between the structures using epitaxial deposition at a deposition pressure of about 700 Torr to about 800 Torr.

14. The processing system of claim 13, wherein the plasma pre-clean comprises a pre-clean temperature of about 100° C. to about 300° C.

15. The processing system of claim 13, further comprising transferring the substrate.

16. The processing system of claim 13, wherein: the plasma pre-cleaning occurs in a first processing chamber of a substrate processing system; and the epitaxial deposition occurs in a second processing chamber of the substrate processing system.

17. The processing system of claim 16, further comprising: transferring the substrate from the first processing chamber to the second processing chamber.

18. The processing system of claim 17, wherein a transfer chamber between the first processing chamber and the second processing chamber has a pressure of about 700 Torr to about 800 Torr.

19. The processing system of claim 13, further comprising: patterning a substrate in a processing chamber prior to the plasma pre-cleaning.

20. The processing system of claim 19, further comprising: the patterning occurs in a first processing chamber; transferring the substrate from the first processing chamber to the second processing chamber; the plasma pre-cleaning occurs in a second processing chamber; transferring the substrate from the second processing chamber to a third processing chamber; and the epitaxial deposition occurs in a third processing chamber.
